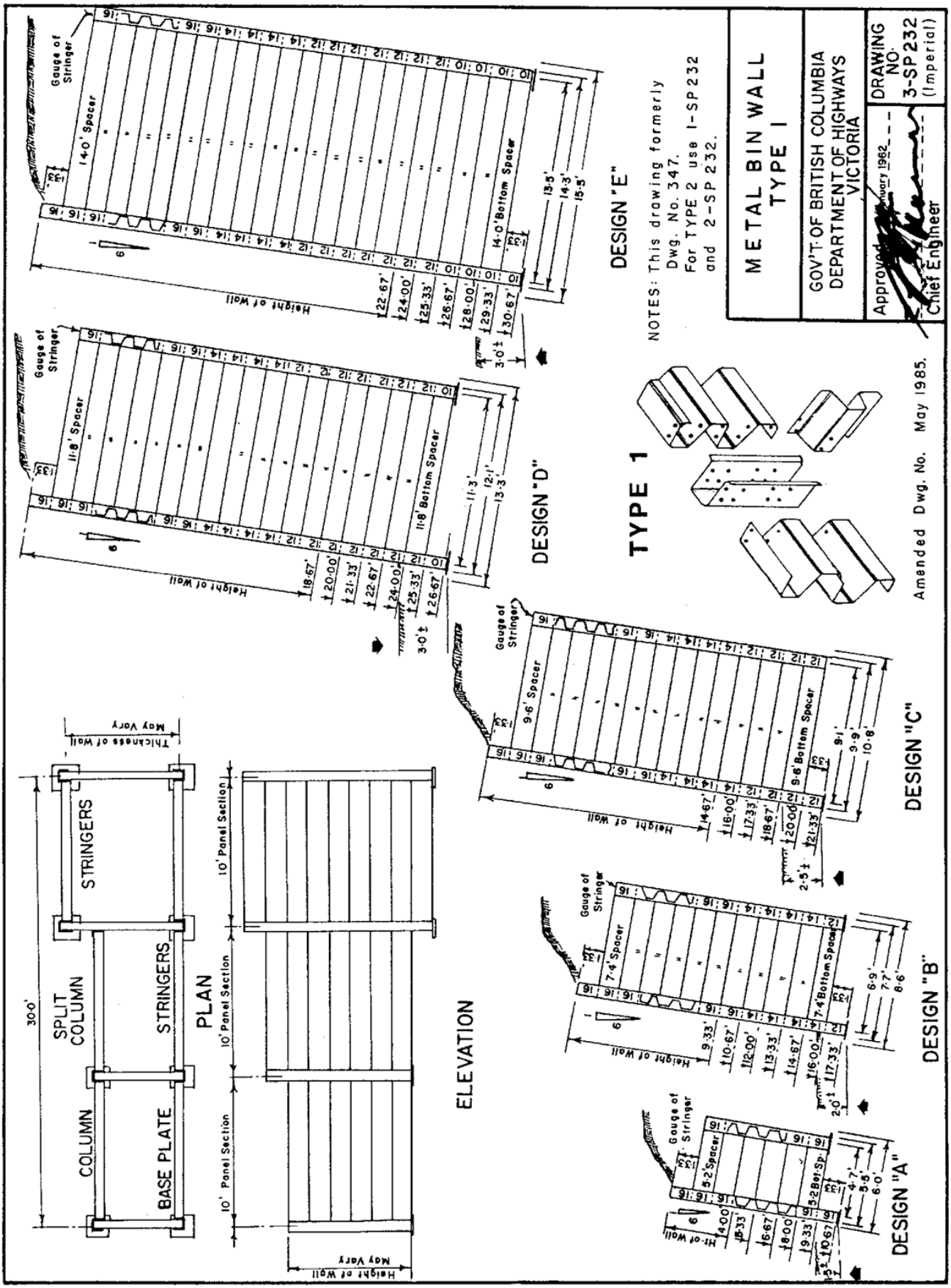

METAL BIN WALL
TYPE 1
GOV'T OF BRITISH COLUMBIA
DEPARTMENT OF HIGHWAYS
VICTORIA
DRAWING NO. 3-SP232 (Imperial)
Approved January 1962
Chief Engineer
NOTES: This drawing formerly Dwg. No. 347.
For TYPE 2 use 1-SP232 and 2-SP 232.
Amended Dwg. No. May 1985.
TYPE 1
DESIGN "E"
DESIGN "D"
DESIGN "C"
DESIGN "B"
DESIGN "A"
ELEVATION
PLAN
SPLIT COLUMN
STRINGERS
COLUMN
BASE PLATE
10' Panel Section
Gauge of Stringer
Height of Wall
Thickness of Wall
May Vary

SECTION 303
CULVERTS

SCOPE

303.01 Scope – This section covers the construction of pipe and box culverts and their associated end treatment. Culverts installed using trenchless techniques are not covered in these Standard Specifications.

303.01.01 Definitions – For the purposes of this Section, the following definitions apply:

(a) Backfill Zones: (See also Figure 303-1)

(i) Backfilling means the operation of filling the embedment and backfill material.

(ii) Backfill Material means fill material used above the embedment material and below the subgrade.

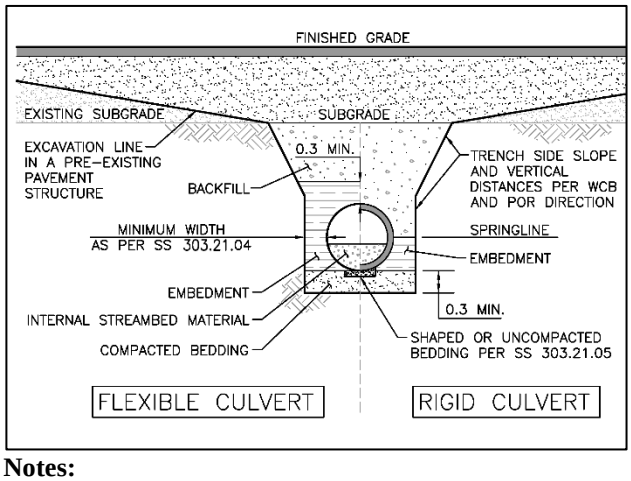
(iii) Bedding Material means the material used to support the culvert.

(iv) Embedment Material means material supporting the culvert, from the top of the bedding to the bottom of the backfill.

(v) Internal Streambed Material means granular material laid inside a culvert to mimic a natural stream bed and aid in fish passage.

(vi) Subgrade means the grade upon which the first layer of granular pavement structure is laid.

Figure 303-1: Culvert Backfill in Trench Installation



Notes:

- "POR" means "Professional of Record"; typically a geotechnical engineer in this case;
- "Rigid" culverts are concrete; all other material types are considered "flexible".

(b) End Treatment Components:

(i) Apron means a horizontal slab in the channel immediately upstream or downstream of an endwall.

(ii) Cut-off Wall means a buried vertical wall below the springline of a pipe to prevent water flow along the exterior of the pipe.

(iii) Endwall means a vertical or sub-vertical wall surrounding the inlet or outlet end of a culvert. Also known as a "headwall".

(iv) Low-Permeability End Seal means installing a piping-resistant material near the end of a pipe to limit inflow of water.

(v) Wingwall means a wall extending to the side of an endwall.

MATERIALS

303.10 Materials – The Contractor shall supply all material in accordance with SS 145.15 and the following:

303.10.01 Concrete Products – supplied in accordance with the Design and the following, as appropriate to the application:

(a) Concrete Pipe (CP) – In accordance with CSA A257, ASTM C14, ASTM C76, ASTM C506, ASTM C507, or ASTM C655.

(b) Concrete Boxes (CBX) – In accordance with ASTM C1433 or ASTM C1577.

(c) Gaskets – Rubber gaskets in accordance with CSA A257.3, ASTM C443, or ASTM C1677.

(d) Flexible Joint Sealants – Flexible joint sealants shall be in accordance with ASTM C990.

Note: Flexible joint sealants shall be used only where explicitly specified by the Designer and only for structures which do not convey water, such as for pedestrian/cyclist tunnels, farm access tunnels or wildlife tunnels.

(e) Concrete Sulphate Resistance – Concrete shall meet the sulphate resistance requirements as stated in the Special Provisions or on the Drawings.

303.10.02 Corrugated Steel Pipe (CSP) – supplied in accordance with SS 320.

303.10.03 Structural Plate Corrugated Steel Pipe (SPCSP) – supplied in accordance with SS 320.

303.10.04 Corrugated Polyethylene Pipe Plastic (PP) – supplied in accordance with SS 317.

303.10.05 PVC Pipe – supplied in accordance with SS 318.03.

303.11 Design Live Load - Culvert products shall be designed for the highway Design live load as follows:

(a) The Design live load in accordance with the Ministry Supplement to the Canadian Highway Bridge Design Code (CHBDC) S6;

(b) AASHTO LRFD Bridge Design Specifications HL-93 live load; or

(c) AASHTO Standard Specifications for Highway Bridges H-20 or HS-20 live load.

303.12 Self-Consolidating Concrete – Self-Consolidating Concrete (SCC) may be used in the manufacture of concrete pipe (reinforced or plain) and box culverts, only under the following conditions.

303.12.01 Products – Only for products limited to the following sizes and specifications:

(a) pipes and barrels smaller than 3.0 m in any internal cross-sectional dimension manufactured in accordance with CSA A257, ASTM C14, C76, C478, C506, C507, or C655.

(b) boxes with both internal spans and rise less than 3.0 m manufactured in accordance with ASTM C1433 or C1577.

303.12.02 Documentation – the Contractor shall provide the following documentation for review upon request by the Ministry Representative:

(a) a full copy of the applicable concrete mix design utilized in the manufacture of any product(s) intended to be incorporated into Ministry works, and

(b) Documentation in accordance with either of the following:

(i) Details in accordance with CSA A23.2-24C documenting the proven history of successful performance for the required strength, durability, and other performance requirements of the proposed mix design for the manufacture of any product(s) intended to be incorporated into Ministry works, or

(ii) A full-scale test shall be used to verify the self-consolidating characteristics for placement and for the hardened concrete properties of the mix design for the manufacture of any product(s) intended to be incorporated into Ministry works. Documentation in accordance with CSA A23.2-24C shall be submitted demonstrating that the proposed mix design will achieve the required strength, durability, and performance requirements.

303.12.03 SCC Mix Requirements – The self-consolidating concrete mix shall conform to the requirements of CSA A23.1, including Clause 8.6 and Table 22, and the appropriate CSA and ASTM standards listed above. The SCC shall meet the following requirements:

(a) Minimum specified compressive strength of 35 MPa at 28 days

(b) S1 class of exposure with the addition of chloride exposure

(c) Exposed to cycles of freeze/thaw

(d) Maximum water-to-cementitious materials ratio (w/cm) of 0.40

(e) Rapid chloride permeability test result (ASTM C1202) of less than 1000 coulombs within 91 days

303.13 Manufacture of Precast Concrete Products

303.13.01 Certified Plants – For Contracts tendered after December 31, 2021, all concrete pipe, boxes, precast concrete endwalls and other precast concrete products shall be supplied from a manufacturer that is certified by an independent third-party certification agency to produce these products according to applicable CSA and ASTM manufacturing standards.

The quality assurance programs recognized specifically for certification of the manufacturer are as follows:

(a) The Canadian Precast Concrete Quality Assurance (CPCQA) Certification Program.

(b) The CSA Group Testing & Certification Inc. (CSA) Certification Program.

(c) QCAST Certification Program by the American Concrete Pipe Association for Precast Products.

(d) National Precast Concrete Association (NPCA) Plant Certification.

(e) Other certification organizations acceptable to the Ministry Representative that are accredited by the Standards Council of Canada for the certification of manufacturers for these products.

303.13.02 Product Marking – All products shall be marked with the certification body's logo confirming that the production of the product is in accordance with the quality and requirements of the specified standards for the products.

CONSTRUCTION

303.20 Construction

303.20.01 Backfill Material – Construction of backfill material shall conform to SS 201 Roadway and Drainage Excavation.

303.20.02 Freezing Temperatures – When the air temperature is below 0°C, no backfilling is allowed unless otherwise accepted by the Ministry Representative. If acceptance is granted, all backfill materials shall be in a thawed state when placed and compacted. Frozen granular backfill materials will not be permitted. No backfill material will be permitted to be placed directly on frozen substrate.

303.20.03 Distortion – During construction of bedding, embedment, and backfilling, the culvert distortion shall be monitored and controlled so that:

(a) the vertical distortion never exceeds 2% of the rise; and

(b) the lateral distortion never exceeds 2% of the span.

303.20.04 Low-Permeability End Seal – Where a culvert has no endwall or cut-off wall, and bedding or embedment is 25 mm IGB or other permeable material, the upstream end of the embankment shall be sealed to minimize the potential for piping failure. Any such seal shall be considered as an End Treatment.

The seal shall be constructed by replacing the IGB/permeable material with 25 mm WGB, a clay seal, or other low permeability, erosion resistant material approved by the Ministry Representative, in the upstream end of the bedding and embedment to a horizontal distance of 1 m into the embankment (see Figure 303-2) and, where shown on the Drawings, may also extend upstream of the inlet.

If used for an end seal, clay shall have plasticity properties lying within the shaded area of Figure 303-3 or be approved by the Ministry Representative.

303.20.05 End Bevels – Any culvert 1000 mm or less in diameter that ends within the "clear zone", as shown on the Drawings, and which is not protected by roadside barrier shall be beveled (mitered) to match the embankment slope, as approved by the Ministry Representative.

303.21 Concrete, Corrugated Steel, Polyethylene and PVC Pipes and Concrete Boxes (this subsection does not cover SPCSP).

303.21.01 Layout – All culverts shall be laid out and constructed in general accordance with the lines, grades, and locations specified in the Drawings, or as directed by the Ministry Representative.

Figure 303-2: Low Permeability End Seal

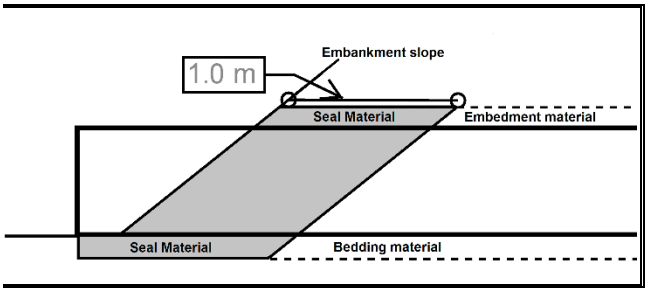
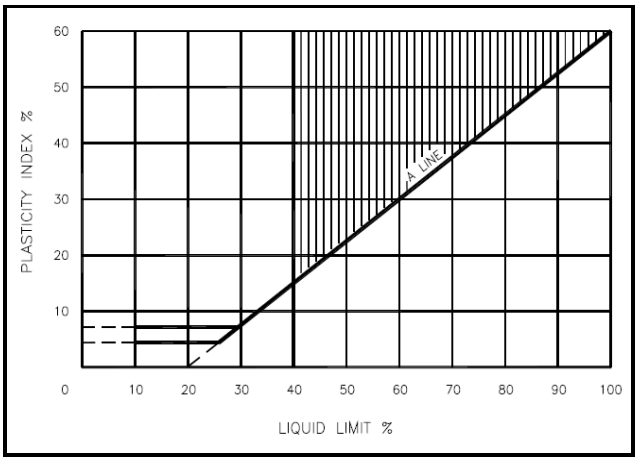


Figure 303-3: Clay End Seal Plasticity Chart



Culverts are to be field fit by the Contractor, to reflect actual conditions encountered on-Site, at a desirable grade of approximately 2% and minimum grade of 0.5%, all to be approved by the Ministry Representative prior to installation.

Such field fits will normally involve minor changes in location, elevation, grade, skew, depth and/or length resulting from actual local drainage course locations, post-stripping surface elevations, final ditch depths, and other factors. Where specified in the Contract or by the Ministry Representative, the field fit may also include adding camber to accommodate any anticipated post-construction consolidation of the embankment.

Some culverts may be designed to control drainage and are not to be field fit without the approval of the Ministry Representative. Any such culverts will be identified in advance for the Contractor, through a note on the Drawings, in the Special Provisions, or by the Ministry Representative.

303.21.02 Flow Obstructions – Any obstacles to flow, such as filter cloth used for siltation control, shall not be place directly on the end of any culvert. Where practicable, there shall be a minimum spacing of 1 m between the end of the culvert and any obstacles to flow.

Any obstructed culvert shall be cleaned out in accordance with SS 165.10.07.

303.21.03 Inlet and Outlet Ditches – Inlet and outlet ditches to culverts shall be constructed to the lines and grades as shown in the Design and field fit as approved by the Ministry Representative.

303.21.04 Trenching – The trench and other preparatory work shall be approved by the Ministry Representative before actual placing starts.

(a) A full trench condition shall be provided wherever possible; a minimum trench depth shall be 50% of the culvert diameter.

(b) The trench width shall be as required to accommodate the Contractor's compaction method and equipment, but not less than 0.3 m on each side of the culvert.

(c) If, in the opinion of the Ministry Representative, the material in the bottom of the excavation is of such a character as to cause unequal settlement along the length of the culvert, the trench shall be dug below the grade to such depth as ordered, backfilled with crushed base course (IGB or WGB) as per SS 202 or other suitable material approved by the Ministry Representative, and compacted as per SS 202 to ensure a firm and uniform foundation.

(d) Where a culvert trench is to be excavated through an existing pavement structure:

(i) the cut slope through the sub-base and base gravels shall be 6H:1V, or as directed by the Ministry Representative, to minimize differential settlement. See Figure 303-1; and

(ii) the road base shall be reconstructed to match the pre-existing pavement structure in material types and thicknesses.

303.21.05 Bedding

(a) Bedding Material – The bedding of all culverts shall consist of:

(i) 25 mm intermediate or well graded base course (IGB or WGB) as per SS 202; or

(ii) other suitable material approved by the Ministry Representative,

and, except as noted in SS 303.21.05(b)(ii), shall be compacted as per SS 202 to 95% Standard Proctor maximum dry density as determined in accordance with ASTM D698, to ensure a firm and uniform foundation.

Where IGB is used, a low permeability end seal shall be installed per SS 303.20.04.

(b) Bedding Construction – Bedding for round or other curved shaped culverts shall be installed to a minimum depth of 300 mm below the culvert, a minimum of 300 mm each side of the culvert, and shall:

(i) have its top surface shaped so that at least 25% of the circumference of the culvert is in contact with the prepared bedding for the whole of its length, or

(ii) be left uncompacted below the middle 1/3 of the pipe, for a depth equal to

(A) for CSP and PP, twice the corrugation depth, or

(B) for PE, PVC, and concrete culverts, 75 mm or 1/24th of the pipe diameter, whichever is greater.

Bedding for Concrete Boxes shall be a minimum of 50 mm in depth (minimum 100 mm over a rock foundation), laid true to grade, and be compacted to a minimum of 95% Standard Proctor maximum dry density as determined in accordance with ASTM D698.

For any gasketed or joint sealed culvert, a transverse trench shall be dug across the bedding at each joint to prevent any granular material from contaminating the joint or impeding the joining process.

303.21.06 Placing – Concrete culvert (CP and CBX) shall be laid beginning at the downstream end with the spigot end pointing downstream. Culvert with elliptical reinforcement shall be laid with the minor axis of the reinforcement as marked by the manufacturer in a vertical position.

Concrete culvert joints shall be sealed using a rubber gasket, installed in accordance with the culvert manufacturer's recommendations.

CSP shall be laid beginning at the downstream end, with separate sections shall be firmly jointed together by a culvert collar.

Where practicable, the first length of culvert at the inlet and outlet shall be full length, to minimize the likelihood of uplift failure.

Corrugated Polyethylene Plastic Pipe with bell and spigot connections shall be laid in the direction with the bell end of the plastic pipe pointing in the opposite direction of water flow, or facing upstream.

Any cuts in CSP shall be coated with two heavy coats of zinc rich paint selected from the Ministry's Recognized Products List under the category of "Additional Paint Coatings – Zinc-Rich Touch-up Paints and Primers".

303.21.07 Embedment – Embedment Material shall consist of 25 mm intermediate or well graded base course (IGB or WGB) as per SS 202, or other mineral aggregate acceptable to the Ministry Representative.

Where IGB is used, a low permeability end seal shall be installed per SS 303.20.04.

Embedment material for embankment installation shall extend a minimum of two-thirds of the span or 0.9 m beyond the culvert span on each side, whichever is greater. Embedment material for trench installation shall extend a minimum of 0.3 m on either side.

Embedment material shall be placed, in layers not exceeding 150 mm in depth when compacted, to an elevation of:

(a) for CSP, PP, PE or PVC culvert, 300 mm or the1/6th of the pipe diameter, whichever is greater, above the culvert crown,

(b) for CP culvert, the springline of the culvert (i.e. up to the widest point of the culvert), and

(c) for CBX culvert, the crown of the culvert.

Embedment material shall be compacted to a minimum 95% (100% within 300 mm of subgrade elevation) of the Standard Proctor maximum dry density as determined in accordance with ASTM D698.

303.21.08 Backfilling – Backfilling material shall be placed and compacted outside and above the Embedment Material, up to subgrade elevation, and shall consist of material suitable for embankment construction in accordance with SS 201.

Backfilling shall be done symmetrically. The differential height of backfilling material on either side of the culvert shall not exceed 300 mm.

The material within 450 mm directly above the crown of the culvert shall be laid and compacted as one lift. For a culvert with crown within 300 mm of subgrade, installation methods shall be approved by the Ministry Representative.

303.22 Structural Plate Corrugated Steel Pipe/Pipe-Arch Culverts (SPCSP)

303.22.01 Excavation – Excavation shall conform to SS 407 Foundation Excavation. If a coarse boulder material or solid rock is encountered when preparing the bed of the SPCSP, the material shall be excavated to a minimum depth of 300 mm below the SPCSP invert and backfilled with granular material having a maximum size of 75 mm.

303.22.02 Assembling – All bolted plates shall be in contact for the full width and length of the seam lap. The bolts in the valley of each longitudinal seam shall be nearer to the visible edge of the plate than the bolts in the crest. The torque on the bolts prior to backfilling shall be in accordance with CSA S6 or as recommended by the manufacturer in the specific plate assembly drawings. A minimum of 5% randomly selected bolts shall be tested in each longitudinal and circumferential connection. The installation shall not be accepted if 10% or more of tested bolts do not meet the specified torque requirements.

Rotation of the pipe culvert and/or spiralling of the longitudinal seams shall not be permitted.

303.22.03 Bedding, Embedment and Backfilling – Bedding, embedment and backfilling shall be in accordance with SS 303.21 and SS 202 unless otherwise specified in the Drawings or the Contract.

For multiple SPCSP structures, structural backfill shall be provided between adjacent SPCSP structures. The minimum clear spacing between adjacent buried structures shall not be less than 1000 mm or one-tenth of the largest span, which ever is greatest, unless approved otherwise by the Designer and the Ministry Representative.

Backfilling shall be done symmetrically. The differential height of backfilling material on either side of the SPCSP at any transverse section shall not exceed 400 mm.

Only hand held compaction equipment is allowed within 1 m behind any endwall or wing wall.End dumped or loose pushed material shall not be piled closer than 3 m from the culvert. Hauling equipment shall not be operated over the culvert until backfilling operations have completed a suitable cover approved by the Ministry Representative.

303.23 Culvert End Treatment – Such treatments may include endwalls, wingwalls, cut-off walls, aprons, traversable grates, up-lift prevention measures, or any combination thereof, and shall be constructed in accordance with the details shown on the applicable Drawings or SP Drawings.

Construction and materials for concrete and reinforcement shall conform to SS 211 and SS 412 and excavation shall conform to the requirements of SS 407.

Rip-rap end treatment is covered under SS 205 and is not included within the Work under this SS 303.

If specified on the Drawings or in the Contract, the Contractor may provide an alternative End Treatment product, as a Value Engineering Proposal in accordance with SS 125. Alternate products shall be designed by a professional engineer registered with Engineers and Geoscientists BC. Sealed engineered drawings of the alternate products shall be provided to the Ministry Representative for review and approval at least two (2) weeks prior to the start of the fabrication of the product.

MEASUREMENT

303.80 Culverts – Culverts will be measured by the METRE along their invert length as installed for each size and type of culvert.

303.81 End Treatments – End Treatments will be measured by EACH for each culvert End Treatment location. Each stand-alone endwall, cut-off wall, wing wall, apron, traversable grate, up-lift prevention measures, or any combination of those components at a single location, will be counted as a single unit**.**

PAYMENT

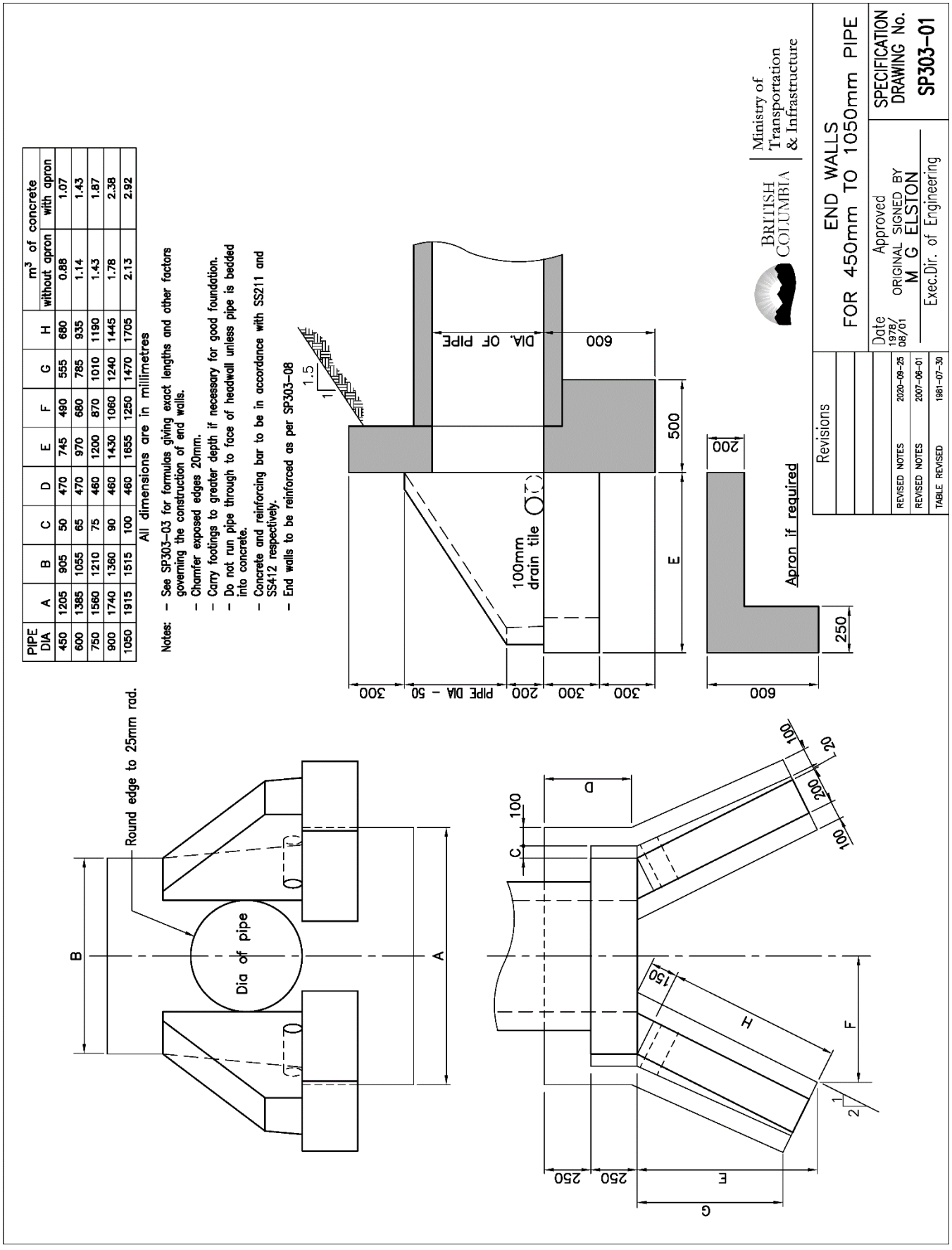
303.90 Culverts – Payment for excavation, culvert supply, assembling, placing, beveling within the clear zone, and backfilling shall be at the Contract Unit or Lump Sum Price bid for each of these items of work, or at the Contract Unit Price bid per metre of culvert.

303.91 Culvert End Treatment – Payment for CULVERT END TREATMENT will be at the Contract Unit or Lump Sum Price bid per size and scope of Culvert End Treatment required.

The Contract Unit Price shall include provision of everything necessary; endwall, (complete with cut-off-wall, wing wall, apron, traversable grate), or up-lift prevention measures, or any combination thereof; incremental costs of materials used as a Low Permeability End Seal; formwork, reinforcement, air entrained concrete, placing and tamping, stripping and cleaning, curing, engineering design for alternate End Treatment products, and any other work necessary in connection therewith, including excavation of Type D material and backfilling, but excluding riprap and Type A excavation.

303.92 Full Compensation – The price(s) bid shall be full compensation for the cost of furnishing all labour, materials, equipment, tools and incidentals necessary to complete the work as specified in the Contract.

303.93 Type A Excavation – Excavation of Type A material encountered in the trench or Culvert End Treatment excavation will be paid at the Contract Unit Price for Type A, or where such an Item is not included in the Contract, as a Change to Work.



PIPE DIA	A	B	C	D	E	F	G	H	m^3 of concrete	
									without apron	with apron
450	1205	905	50	470	745	490	555	680	0.88	1.07
600	1385	1055	65	470	970	680	785	935	1.14	1.43
750	1560	1210	75	460	1200	870	1010	1190	1.43	1.87
900	1740	1360	90	460	1430	1060	1240	1445	1.78	2.38
1050	1915	1515	100	460	1855	1250	1470	1705	2.13	2.92

PIPE DIA	A	B	C	D	E	F	G	H	m^3 of concrete	
									without apron	with apron
1200	2420	1890	115	550	1920	1400	1645	1920	3.51	4.54
1350	2595	2045	125	545	2145	1590	1870	2175	4.07	5.40
1500	2775	2195	140	540	2375	1780	2100	2430	4.60	6.30
1650	2950	2345	150	535	2605	1970	2330	2685	5.50	7.60

All dimensions are in millimetres

Notes:
- See SP303-03 for formulas giving exact lengths and other factors governing the construction of end walls.
- Chamfer exposed edges 20mm.
- Carry footings to greater depth if necessary for good foundation.
- Do not run pipe through to face of headwall unless pipe is bedded into concrete.
- Concrete and reinforcing bar to be in accordance with SS211 and SS412 respectively.
- End walls to be reinforced as per SP303-08

BRITISH COLUMBIA

Ministry of Transportation & Infrastructure

END WALLS FOR 1200mm TO 1650mm PIPE

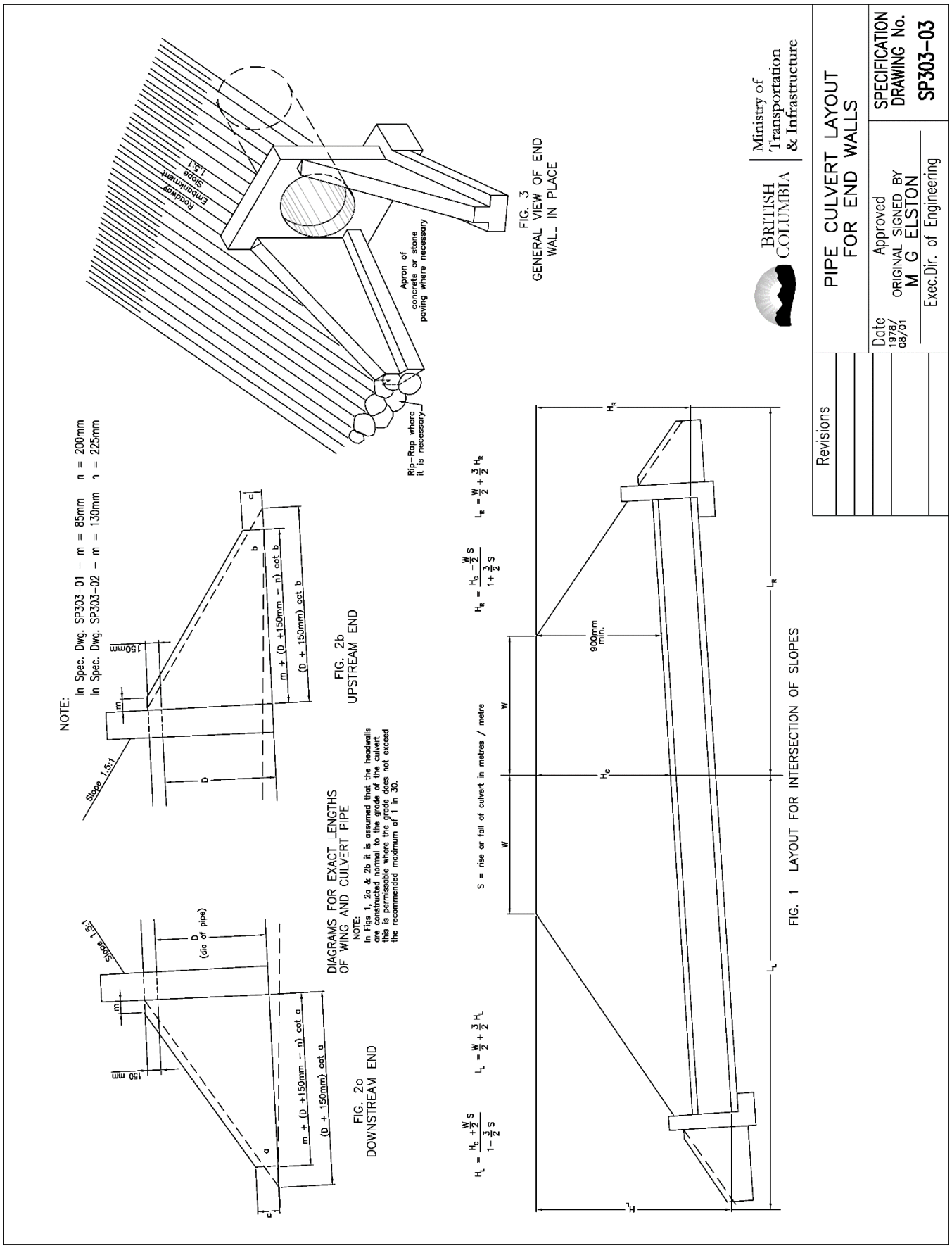
Revisions		
REVISED NOTES	2020-09-25	
REVISED NOTES	2007-06-01	
TABLE REVISED	1981-07-30	

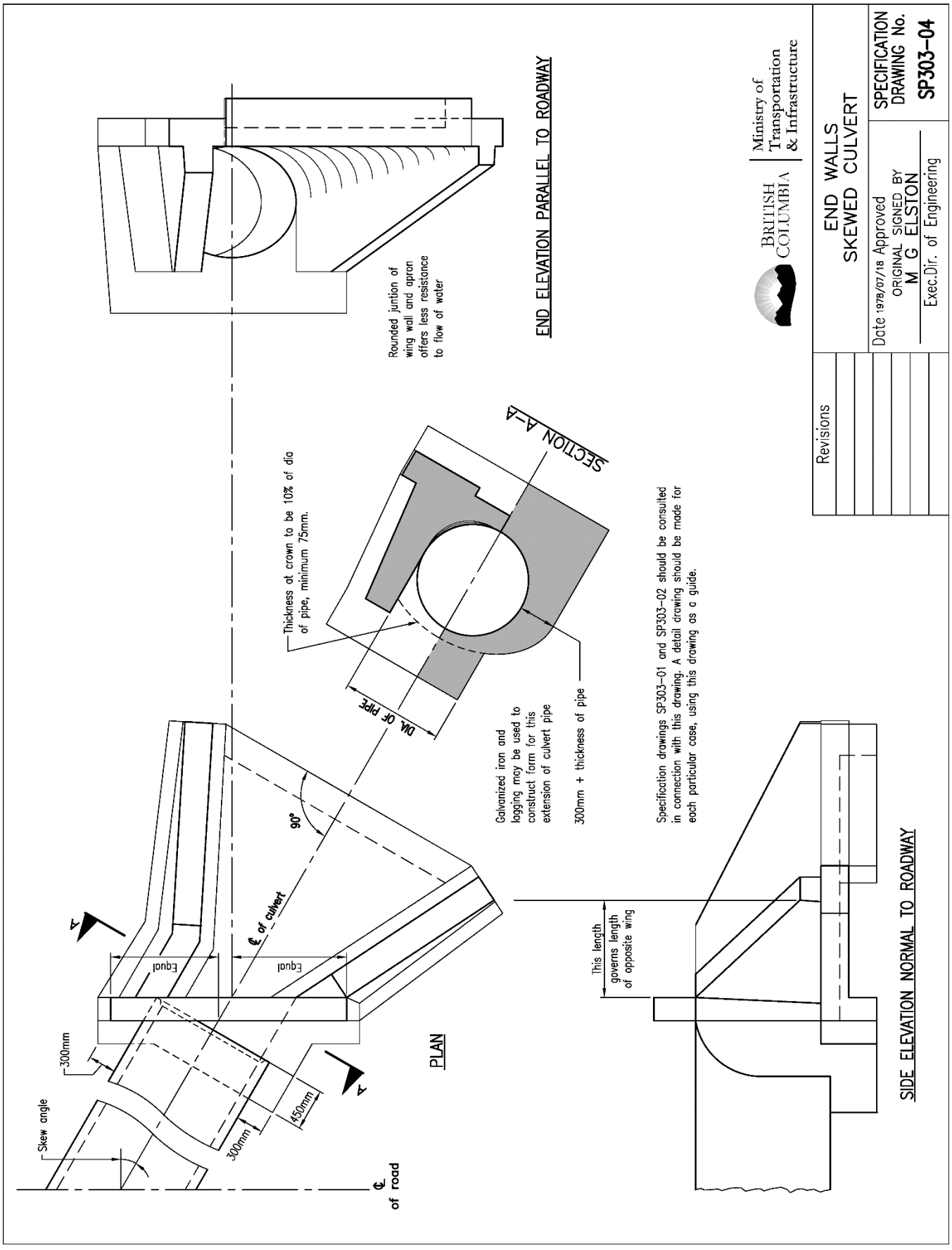
Date 1978/08/01

Approved ORIGINAL SIGNED BY M G ELSTON

Exec.Dir. of Engineering

SPECIFICATION DRAWING No. **SP303-02**


Ministry of Transportation & Infrastructure
BRITISH COLUMBIA
PIPE CULVERT LAYOUT FOR END WALLS
Revisions
Date 1978/08/01
Approved ORIGINAL SIGNED BY M G ELSTON
Exec.Dir. of Engineering
SPECIFICATION DRAWING No. SP303-03
Roadway Embankment Slope 1.5:1
Apron of concrete or stone paving where necessary
Rip-Rap where it is necessary
FIG. 3 GENERAL VIEW OF END WALL IN PLACE
NOTE:
In Spec. Dwg. SP303-01 – m = 85mm n = 200mm
In Spec. Dwg. SP303-02 – m = 130mm n = 225mm
Slope 1.5:1
150mm
m
D
b
c
m + (D +150mm – n) cot b
(D + 150mm) cot b
FIG. 2b UPSTREAM END
Slope 1.5:1
150 mm
m
D (dia of pipe)
a
c
m + (D +150mm – n) cot a
(D + 150mm) cot a
FIG. 2a DOWNSTREAM END
DIAGRAMS FOR EXACT LENGTHS OF WING AND CULVERT PIPE
NOTE:
In Figs 1, 2a & 2b it is assumed that the headwalls are constructed normal to the grade of the culvert this is permissable where the grade does not exceed the recommended maximum of 1 in 30.
S = rise or fall of culvert in metres / metre
W
W
$H_L = \frac{H_c + \frac{W}{2} S}{1 - \frac{3}{2} S}$
$L_L = \frac{W}{2} + \frac{3}{2} H_L$
$H_R = \frac{H_c - \frac{W}{2} S}{1 + \frac{3}{2} S}$
$L_R = \frac{W}{2} + \frac{3}{2} H_R$
900mm min.
H_R
H_c
H_L
L_R
L_L
FIG. 1 LAYOUT FOR INTERSECTION OF SLOPES

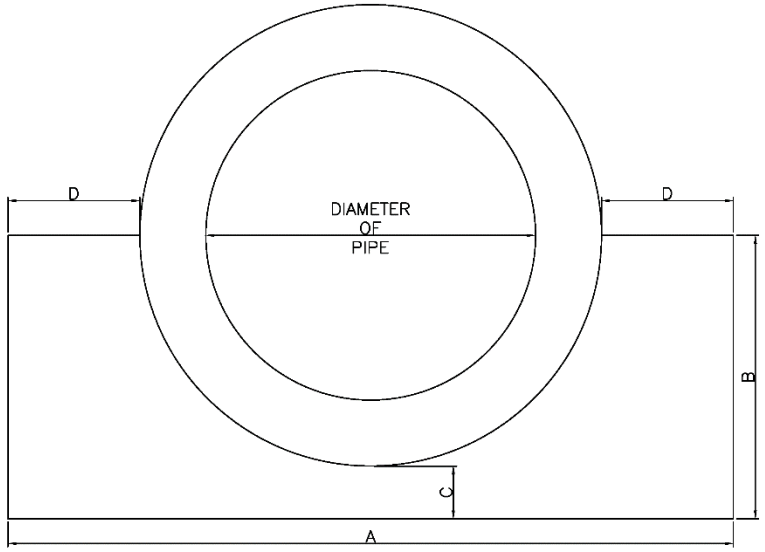

END ELEVATION PARALLEL TO ROADWAY
Rounded juntion of wing wall and apron offers less resistance to flow of water
SECTION A–A
Thickness at crown to be 10% of dia of pipe, minimum 75mm.
DIA OF PIPE
Galvanized iron and logging may be used to construct form for this extension of culvert pipe
300mm + thickness of pipe
Specification drawings SP303–01 and SP303–02 should be consulted in connection with this drawing. A detail drawing should be made for each particular case, using this drawing as a guide.
PLAN
90°
℄ of culvert
Equal
Equal
A
A
300mm
Skew angle
300mm
450mm
℄ of road
This length governs length of opposite wing
SIDE ELEVATION NORMAL TO ROADWAY
BRITISH COLUMBIA
Ministry of Transportation & Infrastructure
END WALLS
SKEWED CULVERT
Revisions
Date 1978/07/18 Approved
ORIGINAL SIGNED BY
M G ELSTON
Exec.Dir. of Engineering
SPECIFICATION DRAWING No.
SP303–04

PIPE DIMENSIONS		CONCRETE CRADLES				
Inside Diameter mm	Wall Thickness mm	A mm	B mm	C mm	D mm	m^3 per m
600	95	1400	500	100	300	0.45
750	108	1580	605	115	300	0.58
900	121	1755	695	115	300	0.69
1050	133	1935	795	130	300	0.84
1200	146	2110	895	140	300	0.99
1350	159	2290	995	150	300	1.16
1500	172	2470	1100	165	300	1.35
1650	184	2645	1200	180	300	1.53
1800	197	2825	1300	190	300	1.73

Notes:
Lower part of cradle to be a stiff mix of concrete, lay pipe to established line and grade.
Follow immediately with concrete of a workable consistency to the required height.
Cradles of these dimensions are for concrete pipes only.

Use concrete cradles under the following conditions.
High fill.
Steep culvert grade.
Rock formations.

Concrete to be in accordance with SS211.



CONCRETE CRADLE

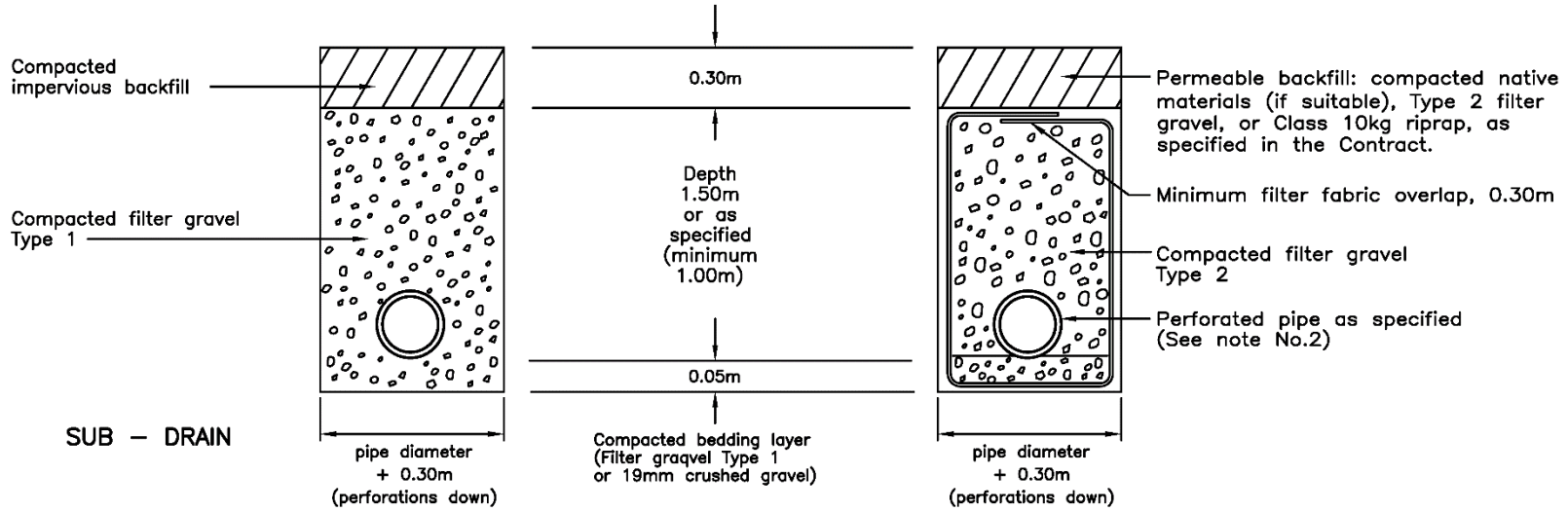
BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

Revisions		
REVISED NOTES	2020-09-25	
TABLE REVISED	1981-07-30	

CONCRETE CRADLE FOR CULVERT PIPE

Date 1978/07/18
Approved
ORIGINAL SIGNED BY
M G ELSTON
Exec.Dir. of Engineering

SPECIFICATION DRAWING No.
SP303-05



SUB – DRAIN

SUB – DRAIN WITH FILTER FABRIC

GENERAL

1. FILTER GRAVEL GRADATION LIMITS

SIEVE SIZE (mm)	PERCENTAGE PASSING	
	TYPE 1	TYPE 2
37.5		100
25.0	100	65–100
12.5	–	20–100
9.5	70–100	0–70
4.75	55–100	0–20
1.18	25–80	0–10
0.30	5–30	0–5
0.075	0–5	0–2

2. MINIMUM PIPE GRADE 0.25%

3. PROVIDE ACCESS STUBS AT:

a) 75m intervals when grade is 0.25% to 1%
b) 100m intervals when grade is 1% to 2%
c) 125mm intervals when grade is over 2%

1. FILTER FABRIC SPECIFICATIONS

i) Minimum Grab Tensile Strength 700N (ASTM D4632)
ii) Minimum Permittivity $1.4sec^{-1}$ (ASTM D4491)
iii) Maximum Apparent Opening Size 0.22mm (ASTM D4751)

2. PIPE ARE REQUIRED WHEN FILTER FABRIC SUB–DRAINS EXCEED:

a) 30m – in clean gravels & sands
b) 50m – in silty sands
c) 100m – in clays

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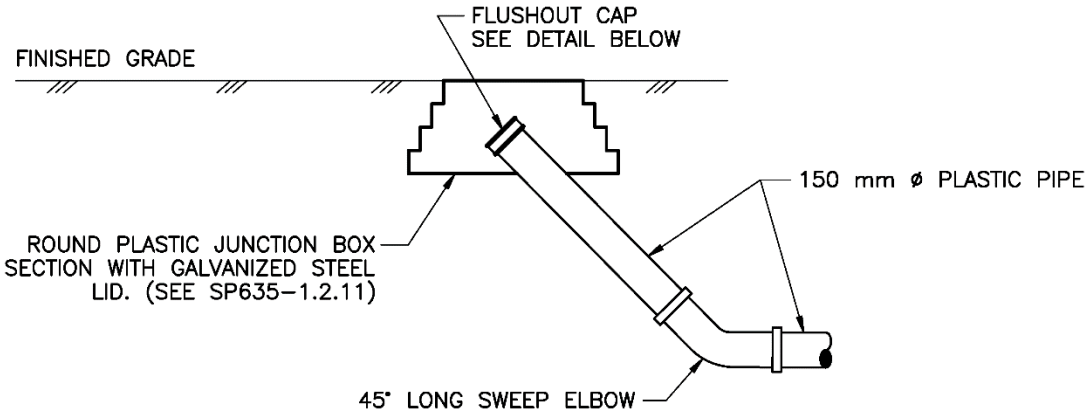
Revisions
REVISED FILTER FABRIC SPEC. –09/2007
REVISED FILTER FABRIC SPEC. –04/2020

SUBDRAIN

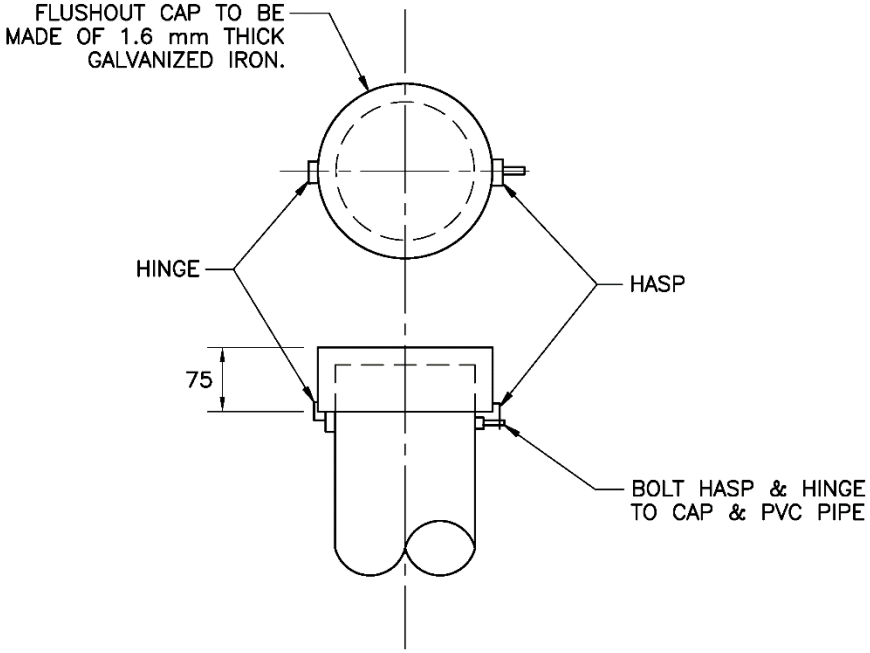
Date 1984/01/18 — Approved ORIGINAL SIGNED BY M G ELSTON, Exec.Dir. of Engineering

SPECIFICATION DRAWING No. SP303–06

SUBDRAIN FLUSHOUT	SP303–07



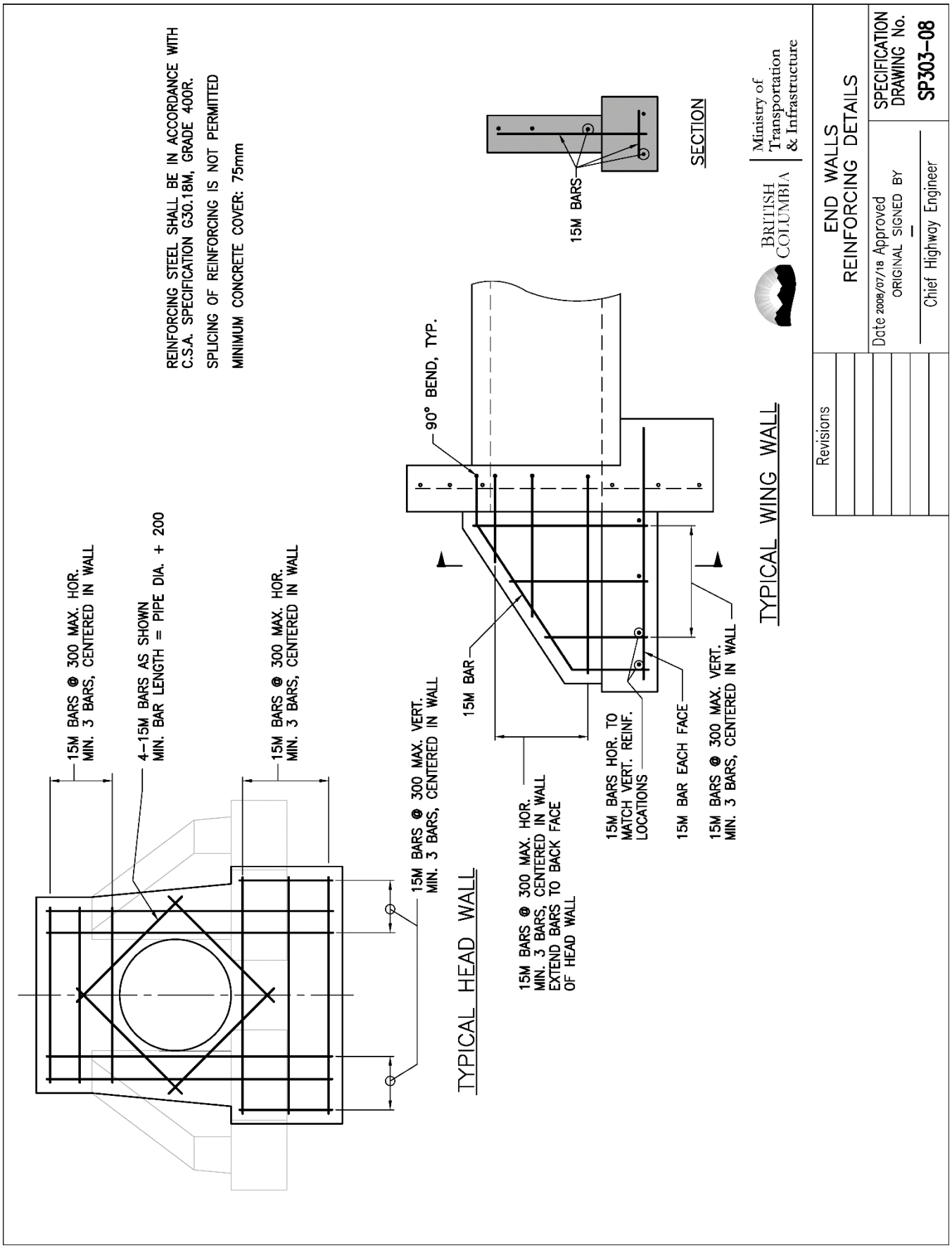
FLUSHOUT AND COVER DETAIL



FLUSHOUT CAP DETAIL

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

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REINFORCING STEEL SHALL BE IN ACCORDANCE WITH C.S.A. SPECIFICATION G30.18M, GRADE 400R.
SPLICING OF REINFORCING IS NOT PERMITTED
MINIMUM CONCRETE COVER: 75mm
15M BARS @ 300 MAX. HOR. MIN. 3 BARS, CENTERED IN WALL
4-15M BARS AS SHOWN MIN. BAR LENGTH = PIPE DIA. + 200
15M BARS @ 300 MAX. HOR. MIN. 3 BARS, CENTERED IN WALL
15M BARS @ 300 MAX. VERT. MIN. 3 BARS, CENTERED IN WALL
TYPICAL HEAD WALL
90° BEND, TYP.
15M BAR
15M BARS @ 300 MAX. HOR. MIN. 3 BARS, CENTERED IN WALL EXTEND BARS TO BACK FACE OF HEAD WALL
15M BARS HOR. TO MATCH VERT. REINF. LOCATIONS
15M BAR EACH FACE
15M BARS @ 300 MAX. VERT. MIN. 3 BARS, CENTERED IN WALL
TYPICAL WING WALL
15M BARS
SECTION
BRITISH COLUMBIA
Ministry of Transportation & Infrastructure
END WALLS
REINFORCING DETAILS
Revisions
Date 2008/07/18 Approved
ORIGINAL SIGNED BY
—
Chief Highway Engineer
SPECIFICATION DRAWING No.
SP303-08

SECTION 308
COATING SYSTEMS

308.01 Scope – This Section defines the coating systems required for shop and field coating of structural steel or wood/concrete surfaces which may be exposed to exterior corrosive conditions.

308.02 Standards and Specifications

308.02.01 Specifying Bodies

(a) AMPP – Association for Materials Protection and Performance, a professional association formed by the merger of NACE International and SSPC in January 2021. Specifications continue to be issued under the NACE and SSPC nomenclature.

(i) NACE – Was the National Association of Corrosion Engineers, also produced specifications and test procedures for coatings and corrosion engineering

(ii) SSPC – Was the Society for Protective Coatings, publishing a large number of coating and cleaning specifications.

(b) ASTM – The American Society for Testing and Materials, produces test procedures for a wide range of products and processes.

308.02.02 Colour Specifications – Using a five-digit number refer to SAE-AMS-STD 595A colour numbers.

308.02.03 Measuring Units – The metric unit (μm) or imperial unit (mil) can be used interchangeably, at the conversion factor of:

1 mil = 25 μm (1 mil = 1 thousandth of an inch)

308.03 System Selection Table – The System Selection Table contained in the Coating Systems section of the Ministry Recognized Products List (RPL) summarizes the coating systems currently in use. The table lists application areas where specified coating systems may be used. It is intended to assist in the choice of a coating system for specific applications. Some systems may find use in more than one specific application or in applications not mentioned.

Where a system is specified in a Contract, only those coating products listed for that coating system may be used.

308.04 System Selection Criteria – The selection of a coating system is dictated by the location and environment of the structure to be coated. The coating system may be applied in a shop or in the field. The field location may necessitate a specific maintenance regimen, for example, a complete recoat or a single overcoat. The environment may be moderately or severely corrosive. In addition, there may be concerns with limiting overspray.

308.04.01 General – The following general comments apply to all coatings.

(a) Coatings shall not be mixed and matched between suppliers, or within a supplier's product line.

(b) All coatings shall bear the manufacturer's name and address, batch number, paint type, date of manufacture, DOH or SAE-AMS-STD 595A color code, and mixing and application instructions.

(c) All labels shall contain the requisite WHMIS label information and be accompanied by a valid MSDS.

(d) All coatings are subject to inspection and testing before acceptance or at the time of application.

308.05 Coating Products – The Ministry Recognized Products List (RPL) lists coating products that have been reviewed and accepted for use on Ministry projects.

The RPL can be reviewed at the following Ministry web site:

https://www2.gov.bc.ca/gov/content/transportation/transportation-infrastructure/engineering-standards-guidelines/recognized-products-list

Products not listed on the Ministry Recognized Products List shall not be used.

Products on the RPL shall be used in accordance with the introduction/definitions, descriptions, notes qualifiers and all other information given in the RPL.

The use of any product from the RPL by the Contractor shall in no way relieve the Contractor from its contractual obligations.

SECTION 312
STEEL TRAFFIC BARRIERS

312.01 Scope – This Section covers the quality and manufacture of steel beam guardrails, wood posts, steel posts, and accessory material for use as traffic barriers.

The layout and installation of materials shall be in accordance the Contract and the SS Drawings of the SP312 series unless otherwise specified.

SS 604 covers the installation of the material described by this Section together with the construction of ancillary work.

Note: SS Drawings numbers SP312-5, SP312-9, and SP312-10 are not used.

312.02 Standards of Reference – Standards of various authorities are referred to herein generally in sufficient detail for the purpose of this Section. For detailed testing and other requirements, copies of the Standards are obtainable as follows:

CSA: Canadian Standards Association:
Standards Sales
13799 Commerce Parkway
Richmond, BC V6V 2N9
https://www.csagroup.org/codes-standards/

ASTM: American Society for Testing and Materials,
1916 Race Street,
Philadelphia, PA 19103
https://www.astm.org/Standard/standards-and-publications.html

TF13: Task Force 13
Guides: https://tf13.org/guides/
Component Search: https://tf13.org/components/

AASHTO: American Association of State Highway and Transportation Officials
444 North Capitol Street N.W., Suite 225
Washington, DC 20001
https://store.transportation.org/

312.03 General – Steel traffic barriers shall consist of rail sections fabricated for installation to develop continuous beam strength with the necessary safety end assemblies.

End assemblies shall be "Manual for Assessing Safety Hardware (MASH)" certified.

All rail sections and other components shall match the design profiles, specifications and dimensions of the Task Force 13 requirements cited below, in part to ensure full interchangeability of similar components regardless of the source of manufacture.

Unless otherwise specified, guardrails and posts and accessories shall be in accordance with SS Drawings SP312-1 SP312-2, SP312-3, SP312-4, SP312-6, SP312-7, SP312-8, and SP312-11. Design and part numbers are taken from the Task Force 13 "Guide to Standardized Roadside Hardware" web resources and are reproduced on these drawings.

Task Force 13 information is available on-line at:

http://www.tf13.org/Guides/index.php

Drawings for individual components may be found by a search from the Task Force 13 "Components" web page:

https://tf13.org/components/

The name and/or trademark of the manufacturer and the metal thickness shall be clearly and permanently marked on each component clear of the splicing overlap and on the opposite face to the traffic side.

312.04 Steel Beam Guardrail – Guardrail shall be manufactured in accordance with AASHTO M 180 "Standard Specification for Corrugated Sheet Steel Beams for Highway Guardrail" and shall conform to the following detail requirements.

312.04.01 Beam Classification (According to the AASHTO Standard) – The shape, class and type, together with the Task Force 13 design and part numbers, shall be as in accordance with the Special Provisions, Contract Drawings and/or SS Drawings of the SP312 series.

(a) Shape – W-Beam are for SGR38 installations, and Thrie-Beam for SGR09 and SGM09 installations and both shall be accurately formed to profile, dimensions and tolerances of AASHTO M 180 with overall cross sections of 312 mm x 85 mm and 508 mm x 85 mm respectively (TF13 Designation RWM01a to RWM22a and RTM02a to RTM06a).

(b) Class and Type – Unless otherwise, beams shall be Class A, Type II.

(c) Length – Normally 3.81 m (12' 6") nominal length beams (4.19 m, +0 mm/-75 mm overall) will be required.

(d) Punchings – In addition to joining bolt holes, punchings may be designed as 7.62 m, 3.81 m, 1.905 m or 0.953 m (25' 0", 12' 6", 6' 3", 3' 1½" respectively), see SS Drawings SP312-1 and SP312-2 for requirements.

(e) Curved Rails – For curved rail data see SS Drawing SP312-11.

312.04.02 Metal

(a) **Properties** – Properties of the base metal for beams and any transition section shall conform to the following requirements:

(i) Yield point minimum: 345 MPa

(ii) Tensile strength minimum: 483 MPa

(iii) Elongation minimum in 50 mm: 12%

Note: End and buffer sections shall have a minimum yield point of 227 MPa and a minimum tensile strength of 310 MPa.

(b) **Sheet Thickness** – Nominal and finished thicknesses shall be in accordance with Table 312-A.

(c) **Sheet Widths** – Sheet Widths shall be minimum 483 mm for W-Beams and 749 mm for Thrie-Beams both with a permissible tolerance of minus 3.2 mm.

312.04.03 Zinc Coating – The zinc used for the coating of Type I and II sections shall be as prescribed in ASTM B6, and shall be at least equal to the grade designated as "Prime Western."

Steel galvanized before beam fabrication shall be coated in accordance with ASTM A653. Beams galvanized after fabrication shall conform to the requirements of ASTM A123. In either case, the coating shall be firmly bonded to the base metal after fabrication is complete.

The average mass of zinc coating Type I (AASHTO M 180) shall be 610 g/m² for three specimens but the minimum is 550 g/m² for any individual specimen. The average mass of zinc coating Type II (AASHTO M 180) shall be 1220 g/m² for three specimens but the minimum is 1100 g/m² for any individual specimen. The mass of coating is the total amount on both sides of the sheet or beam.

312.04.04 Quality of Work – Beams shall be punched for splice and post bolts in strict conformity with AASHTO M 180 to the designated number, and centre to centre spacing, of posts.

Upon delivery the beams shall be ready for assembly and installation.

Field cutting shall be avoided where practicable. Where necessary, field cutting shall be performed with a cut-off or hack saw; field hole cutting shall be performed with a drill and/or reamer. Flame or plasma cutting, or welding, shall not be permitted.

All field cuts or holes shall be deburred and treated with two heavy coats of zinc rich paint selected from the Ministry's Recognized Products List under the category of "Additional Paint Coatings – Zinc-Rich Touch-up Paints and Primers".

Warped or otherwise deformed beams, and those with injurious defects or excessive roughness of the zinc coating will be rejected.

Beams for concave and/or convex curved installations with a radius of 45 m or less shall be accurately and smoothly bent in the shop with minimum deformity. The radius of curvature shall be clearly stencilled on the back of each curved rail section.

312.04.05 Examination – Examination may include the following:

(a) Dimension measurements for size and thickness all of which shall be within the tolerances specified by SS 312.04 and Table 312-A.

(b) Coating generally shall be smooth, free of beading or sharp projections at edges. Coating adherence shall prevent the peeling of any portion of the zinc coating so as to expose the base metal by cutting or prying with a stout knife under considerable pressure.

Table 312-A: Beam Thickness by Class – Base Metal and Finished Galvanized

Class	Base Metal Nominal Thickness (mm)	Finished Galvanized Beam Thickness (mm)		Tolerance (mm)
		Type I	Type II	
A	2.67	2.74	2.82	minus 0.23
B	3.43	3.51	3.58	minus 0.25
Note: Tolerance given is for under specified thickness; there is no limit for over thickness.				

312.04.06 Testing – When required by the Ministry Representative, galvanizing testing shall be carried out according to ASTM A90 or AASHTO T 65 to determine the mass of galvanizing in conformity with the requirements of SS 312.04.03. Alternatively, magnetic gauge determination in accordance with ASTM E376 may be permitted by the Ministry Representative.

The testing of the mechanical properties of the base metal may be carried out by the Ministry Representative for compliance with the requirements of SS 312.04.02.

312.04.07 Rejection and Retests – Should the beam tested fail to meet the testing requirements specified above, two additional beams as selected at random by the Ministry Representative shall both meet the test requirements in every respect, otherwise the complete lot represented by the samples will be subject to rejection.

In the event of complete lot rejection and prior to any further testing by the Ministry, the intended replacement lot shall be satisfactorily tested by a mutually acceptable independent testing agency at no expense to the Ministry.

Rejected material shall be replaced at the Supplier's or Contractor's expense, including shipping charges and

removal of rejected material at the construction site, if applicable.

312.05 Guardrail Accessories

312.05.01 Transition Section – Transition sections for W-Beam to Thrie-Beam shall be of matching metal thickness and finish and be fabricated to the Task Force 13 design profile (TF13 Designation RWT01a-b or RWT02a-b) to provide a smooth and uniform transition of beam widths and corrugations.

312.05.02 Terminal Connectors – For connections to bridge parapets, abutments and the like, TF13 Designation RWE02b or RTE01b terminal connectors shall be used and the finish shall be Type II.

312.05.03 Back-up Plates – Back-up Plates are required where steel beams between splices are to be installed to any steel posts (standard 305 mm lengths) and shall be fabricated of the stipulated lengths from the standard beam section (as per TF13 Designation RWB01a or RTB01a).

312.05.04 Fasteners and fittings – All fasteners and fittings shall be in accordance with TF13 and shall be galvanized in accordance with ASTM A153 or ASTM A123 as appropriate.

312.05.05 Cable Assembly – Cable Assembly for Breakaway Cable Terminal (BCT) installation shall consist of a swaged fitting, 25 mm diameter threaded stud, locking pin, nut and washer at each end of a cable of 19 mm nominal diameter, 6 x 19 wire strand core independent wire rope core, galvanized, right regular lay wire rope with a minimum breaking strength of 200 000 N (45,000 lb) and conforming to AASHTO M 30 with the overall assembly length of 1980 mm developing a similar breaking strength (TF13 Designation FCA01, FWC24a, and complete with plate washers, FPB01).

312.05.06 Anchor Plate – Anchor Plates for the breakaway cable attachment to the steel beam for BCT assemblies shall be fabricated to the TF13 design profile (TF13 Designation FPA01) from steel plate conforming to ASTM A36 and galvanized according to ASTM A123 after forming and drilling for and including eight 16 mm diameter hexagonal-head bolts 38 mm long each with nut and 3.0 mm thick plain washer 44 mm outside diameter (TF13 Designation FBX16a and FWC16a) complying with SS 312.05.04.

312.05.07 Anchor Inserts – Anchor inserts for fixing terminal connector end sections to existing concrete shall be as shown on the Drawings or as directed by the Ministry Representative.

312.05.08 Steel Posts

(a) Concrete Installation – Installation of steel posts on concrete components and the transitions of the barrier system to the concrete components shall be as shown on the Drawings or as directed by the Ministry Representative.

(b) Soil Embedment Installation – Unless otherwise specified, details shall be in accordance with SS Drawings SP312-2, SP312-3, and SP312-4. Steel posts shall be fabricated from standard structural steel H-beam that conforms to the latest version of any of the following standards:

(i) AASHTO M 270 (ASTM A709), Grade 36 [250] or Grade 50S [345S] without the supplementary requirements;

(ii) ASTM A36; or

(iii) ASTM A992.

After the section is cut-to-length and all holes are drilled or punched, each guardrail post shall be hot dip galvanized in accordance with ASTM A123.

Any areas where the galvanizing is damaged during installation shall be touched up with two heavy coats of zinc rich paint selected from the Ministry's Recognized Products List under the category of "Additional Paint Coatings – Zinc-Rich Touch-up Paints and Primers".

312.05.09 Steel Blockouts – Steel blockouts for modified Thrie-beam guardrail shall be manufactured to any of the following standards:

(i) AASHTO M 270 (ASTM A709), Grade 36 [250] or Grade 50S [345S] without the supplementary requirements;

(ii) ASTM A36; or,

(iii) ASTM A992

After the section is cut and all holes are drilled or punched, the components shall be hot dip galvanized in accordance with ASTM A123.

Any areas where the galvanizing is damaged during installation shall be touched up with two heavy coats of zinc rich paint selected from the Ministry's Recognized Products List under the category of "Additional Paint Coatings – Zinc-Rich Touch-up Paints and Primers".

Unless otherwise specified, the blockout type, size and length shall be in accordance with TF13 Designation PWB03, complete with connecting bolts and nuts (FBX16a).

312.06 Treated Wood Posts and Offset Blocks

312.06.01 Rectangular Posts – Unless otherwise specified, details shall be in accordance with SS Drawings SP312-1, SP312-2, and SP312-3.

(a) Wood – Rectangular posts and offset blocks shall be Douglas Fir/Hemlock "No. 1", Structural Posts and Timber, graded in conformity with the requirements of SS 909.

(b) **Wane –** Wanes on any face shall not exceed the following width, being the minimum permissible post width less the portion entirely free of wane:

(i) Above grade (including blocks): 25 mm

(ii) Below grade: 60 mm

(c) **Dimensions –** Posts and blocks shall be supplied in the exact lengths specified and shall be rough-sawn or S4S dressed, nominal 200 mm x 150 mm, with dressed minimum dimension of 189 mm x 138 mm and a tolerance of plus 3 mm.

312.06.02 Preservation Treatment – All required wood components shall be pressure treated in accordance with SS 908.

The preservatives and retention of preservatives for all posts and blocks shall conform to SS 908 from the classification use table under the heading of "Roadside guardrail posts and other guardrail components".

Treated wood shall also meet the requirements of SS 604 - Steel Traffic Barrier Construction.

Ideally, all posts and blocks shall be pressure treated after being cut to the specified length. However, the buried end of any post must be pressure treated in a treatment facility, but the rest of the cuts may be field treated in accordance with SS 908.07.

312.07 Quality Control – The Contractor shall be fully responsible for all Quality Control testing, inspection and documentation to achieve compliance with the Contract.

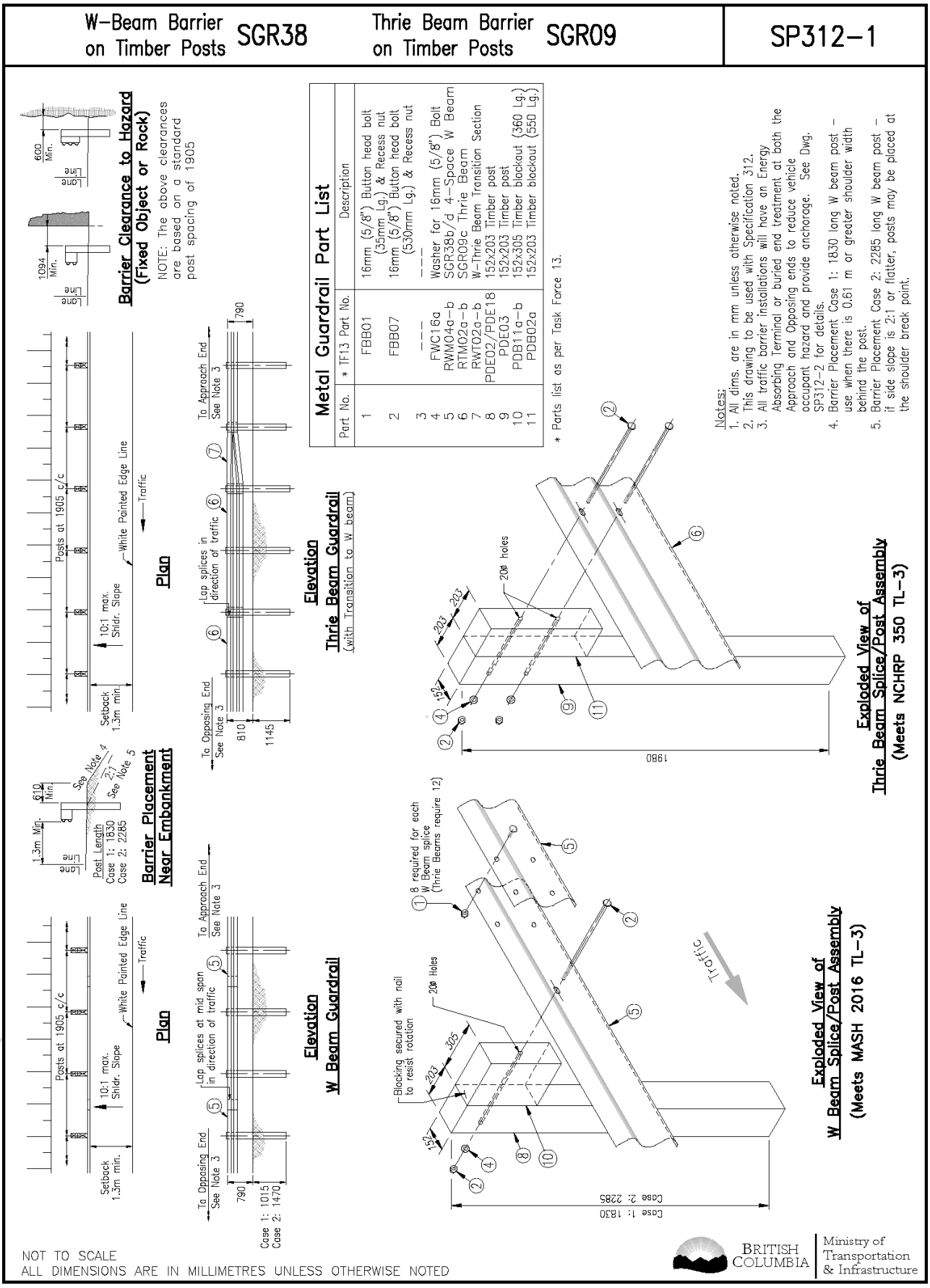
The Contractor shall provide documented test results for the materials to the Ministry Representative for review in accordance with the Contract, and upon request.

312.08 Quality Assurance – The Ministry may implement a Quality Assurance program to confirm compliance with the Contract. Without limitation, the QA activities may include auditing the Contractor's and fabricator's Quality Control program, inspection, and testing, all at the Ministry's discretion.

All materials may be inspected by the Ministry before shipment to the construction site, or at the construction site, or both. The supplier or manufacturer shall provide the necessary facilities to enable the Ministry Representative to expeditiously examine selected pieces or take specimen samples for testing from the material lot(s) clearly identified for intended supply to the Ministry. Identification shall be such that after inspection and testing the lot may be subsequently identified for acceptance or rejection.

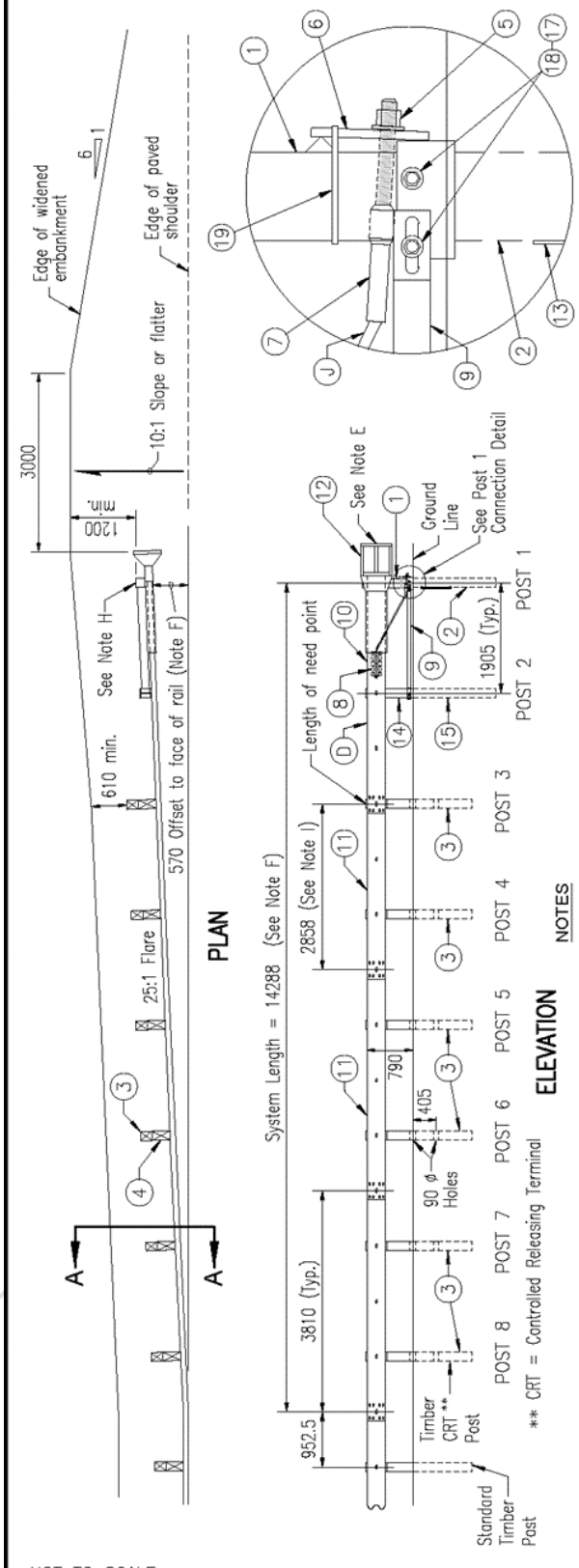
Inspection of material before shipment shall not preclude its subsequent rejection at the construction site if found to fail any requirements of this specification.

W-Beam Barrier on Timber Posts SGR38 | Thrie Beam Barrier on Timber Posts SGR09 | SP312-1



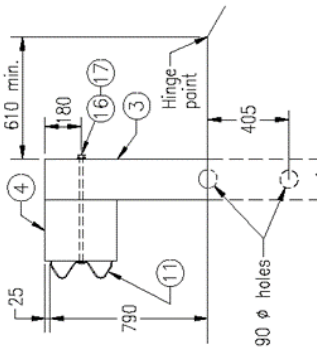
Midwest Guardrail System (MGS)
Energy Absorbing Terminal (Timber/Steel Posts)

SP312–2



POST 1 CONNECTION DETAIL

(See manufacturer's information for additional details)



TYPICAL AT POST No. 3 TO 8

NOTES

A. The terminal shown (MSKT-MGS as manufactured by Road Systems Inc.) is MASH compliant at Test Level Three (TL-3) and is shown for illustration purpose only. It may be used for all posted speeds.

B. Other guardrail terminal systems that meet the Manual for Assessment of Safety Hardware (MASH) criteria and listed on the Recognized Products List are acceptable. Contact the BC MoTI Principal Highway Safety Engineer to request approval to use other systems.

C. The terminal shown and other acceptable MASH compliant (TL-3) end terminal systems shall be installed according to manufacturer's recommendations.

D. Different manufacturers require different perforated W beam end rails. Hardware may vary between manufacturers. See manufacturer's information.

E. A reflectorized object marker shall be installed according to manufacturer's recommendations.

F. Provide an offset up to 570 mm so that the impact head does not encroach onto the paved shoulder. The offset is provided over the length of the terminal system from the center of the last splice to the end of the W beam at Post 1.

G. Posts 3 to 8 are timber Controlled Releasing Terminal posts. Posts beyond the terminal system length are standard timber posts per SP312-1.

H. The W beam end rail panel is not attached to Post 1. Blockouts are not used on Posts 1 and 2.

I. The W beam end rail panel is 3810 long (C/L Post 1 to C/L Post 3). The second rail must be 2858 long (C/L Post 3 to C/L splice) to establish the mid-span splices between posts.

J. After final assembly, recheck cable to be sure it is taut and has not relaxed.

Energy Absorbing Terminal (EAT) Part List

PART NO.	* TF13 PART NO.	DESCRIPTION Materials provided be MGS EAT manufacturer. See manufacturer's details for more information.
1		Upper post No. 1, 152 x 152 x 3.2 Tube
2		Lower post No. 1, W150x22.5
3	PDE09	Timber CRT post, 152 x 203 x 1830 long
4	PDB11a-b	Wood blockout, 152 x 305 x 360 long
5	FNX24a	Nut c/w Washer FWC24a
6	FPB01	Bearing plate
7	FCA01	BCT cable anchor assembly
8	FPA01	Cable anchor box
9		Ground strut
10	RWM04a	Perforated W beam rail end panel
11	RWM04a	Standard W beam rail. Multiple sections required. Sections vary in length.
12		Impact head
13		Soil plate
14		Upper post No. 2, W150x13.5
15		Lower post No. 2, W150x13.5
16	FBB07	16 mm Button head bolt (530 mm Lg.) & recess nut
17	FWC16a	Washer for 16 mm bolt
18	FBX16a	Hex bolt (230 mm Lg.) and nut
19		Bearing plate retainer tie

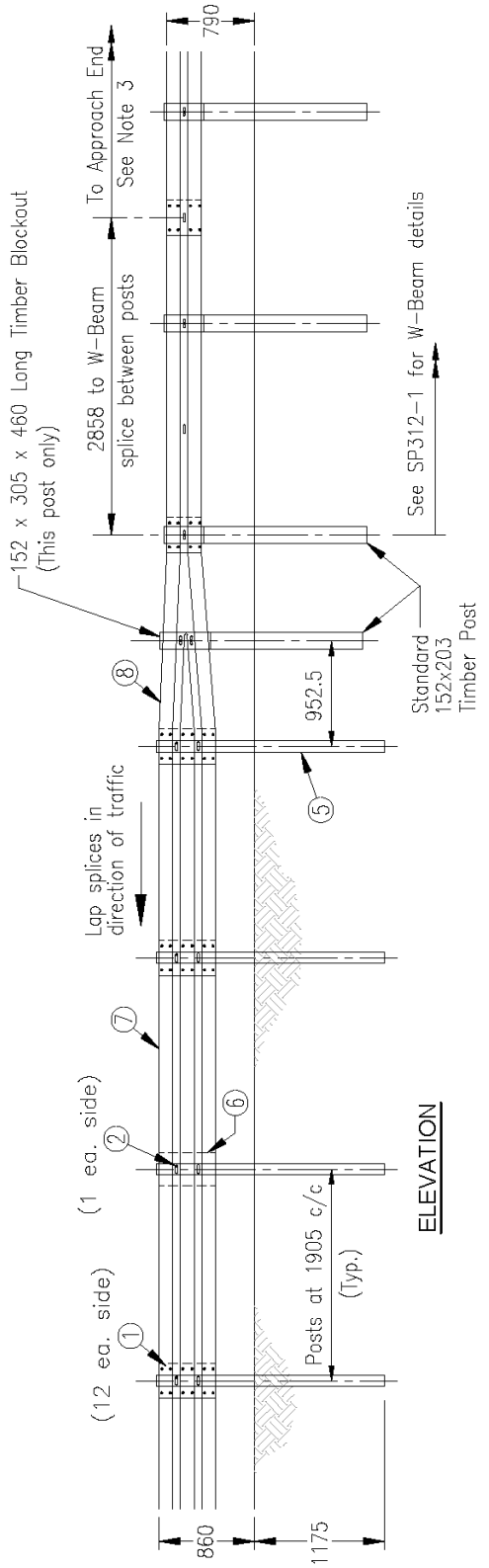
* Parts list as per Task Force 13.

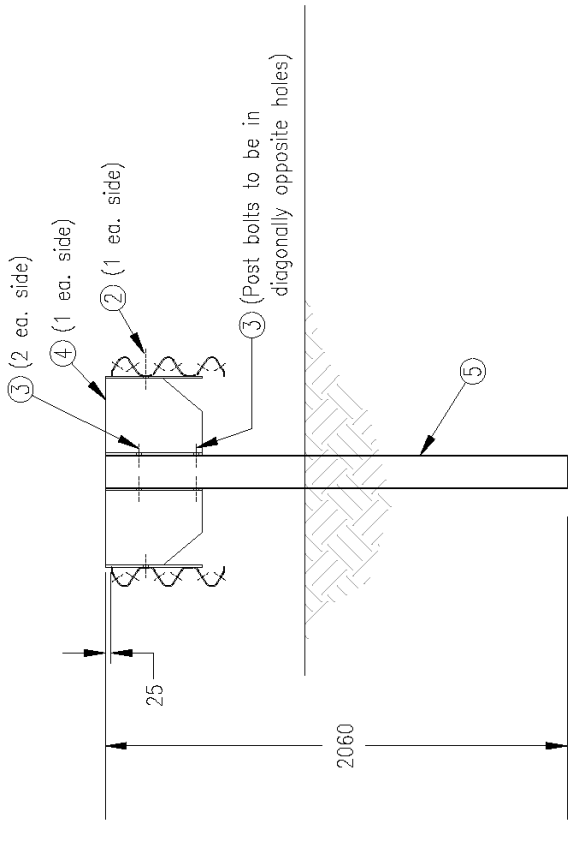
NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

Modified Thrie Beam Barrier on Steel Posts SGM09b

SP312-3





Metal Guardrail Part List

Part No.	* TF13 Part No.	Description
1	FBB01	16mm (5/8") Button head bolt (35mm Lg.) & Recess nut
2	FBB02	16mm (5/8") Button head bolt (50mm Lg.) & Recess nut
3	FBX16a	16mm (5/8") Hex head bolt (40mm Lg.) & nut
4	PWB03	Modified Thrie Beam Blockout
5	PWE04	Steel Guardrail Post W150x13.5
6	RTB01a	Thrie Beam Backup Plate
7	RTM02a	SGM09b Thrie Beam
8	RWT01a	W - Thrie Beam Transition Section

* Parts list as per Task Force 13.

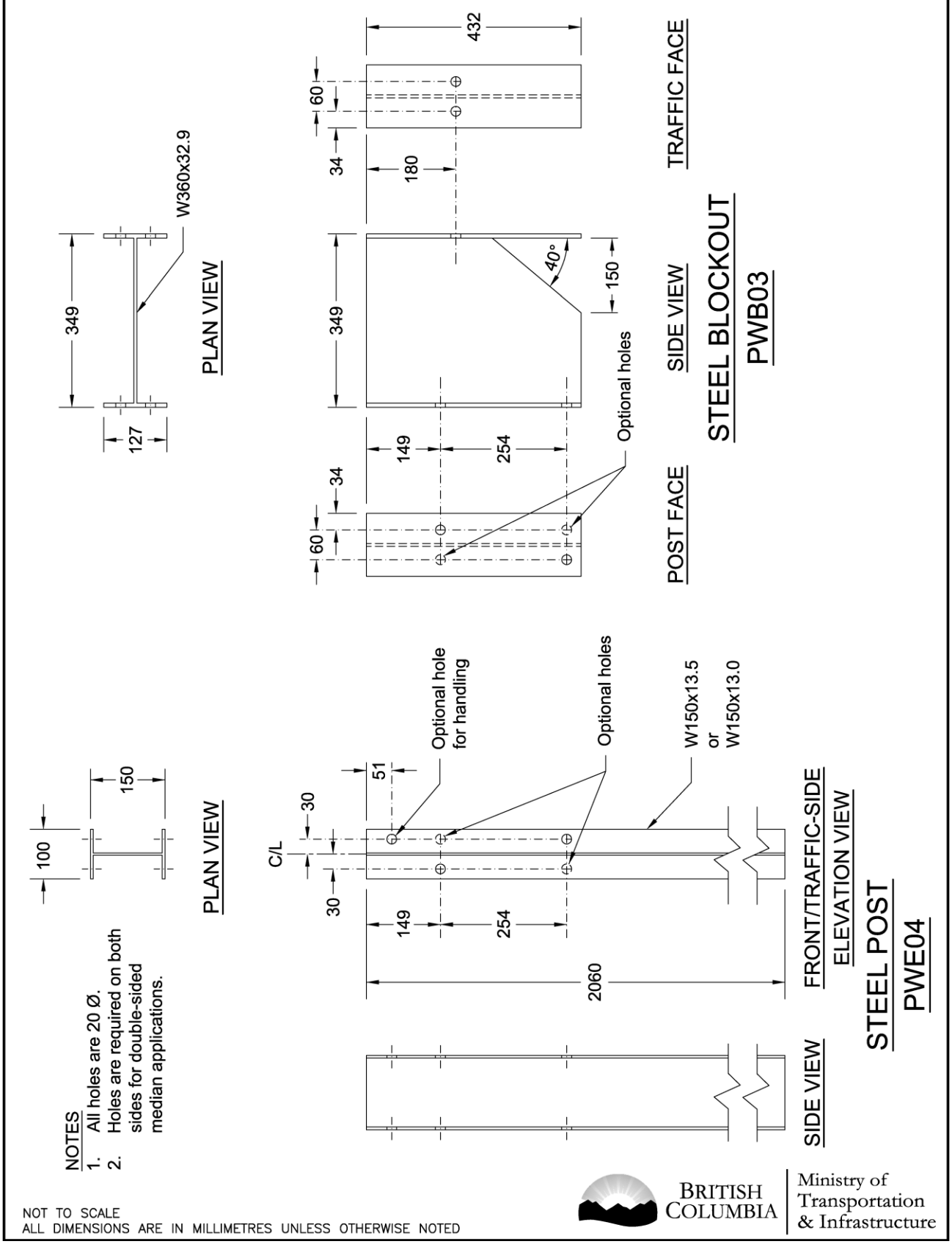
Notes:
1. All dims. are in mm unless otherwise noted.
2. This drawing to be used with Specification 312.
3. All traffic barrier installations will have Energy Absorbing Terminal end treatments at both the Approach and Opposing ends to reduce vehicle occupant hazard and provide anchorage. See Dwg. SP312-2 for details.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

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Modified Thrie Beam Steel Post and Blockout

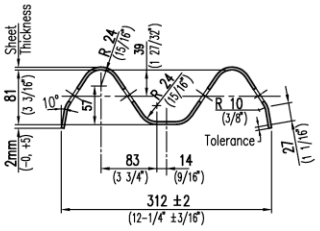
SP312–4

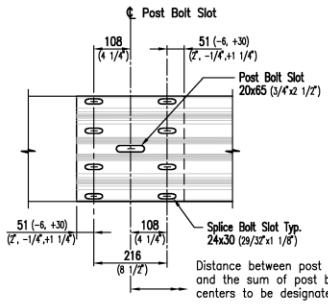


Steel Guardrail Beam Parts

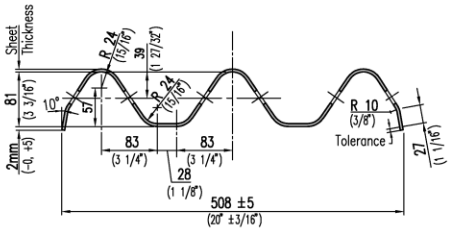
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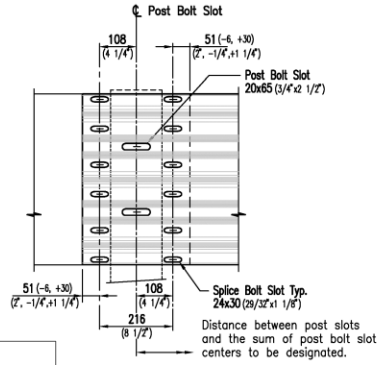
W Beam
RWM01a – 22a



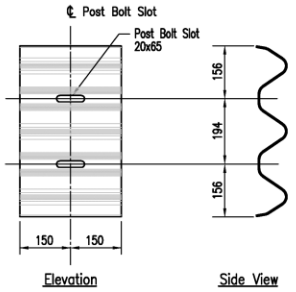


Thrie Beam
RTM02a – 06a

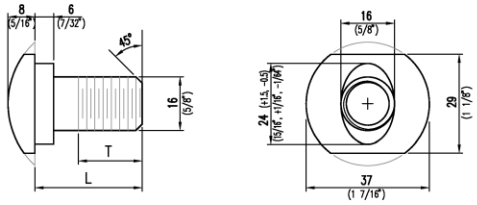




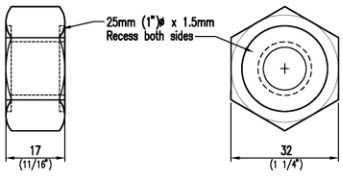
Back–up Plate (non–splice posts)
RTB01a–b



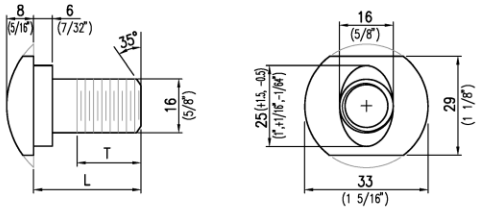
16mm (5/8") Button Head Bolt and Recess Nut
FBB01–05



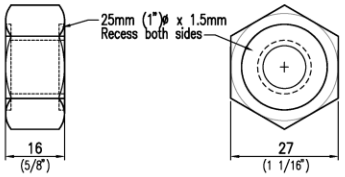
DESIGNATOR	L	T (Min.)
FBB01	35	30
FBB02	50	45
FBB03	255	100
FBB04	460	100
FBB05	640	100



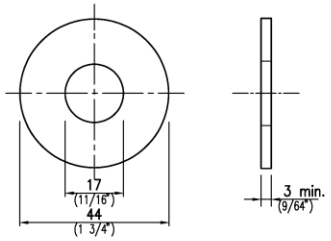
16mm (5/8") Button Head Bolt and Recess Nut
FBB06–07



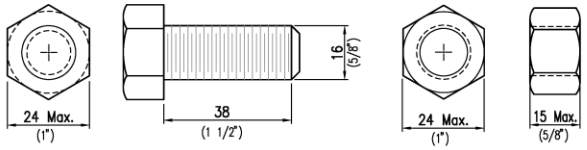
DESIGNATOR	L	T (Min.)
FBB06	360	100
FBB07	530	100



Washer for 16mm (5/8") Bolt
FWC16a



16mm (5/8") Hex Bolt and Nut
FBX16a



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